Huizhen (Jenny) Sun

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Research Interests

Hyperspectral and ocean colour remote sensing, coastal and inland water quality, climate impacts on coastal environments, spatial statistics, and geospatial machine learning.

Education

MSc in Geospatial Data Science Sep 2025 – Aug 2026
The University of Hong Kong *Hong Kong SAR, China*

- GPA: 3.61/4.3 Dissertation supervisor: Prof. Hongsheng Zhang
- Relevant coursework: Satellite Data Processing and Information Extraction; Spatial Statistics; Cloud Computing for Geospatial Data Analytics; Geographic Information Systems

Bachelor of Management in E-Commerce Sep 2021 – Jun 2025
Xidian University *Xi'an, China*

- GPA: 3.6/4.0 (84%)
- Relevant coursework: Probability Theory and Mathematical Statistics; Applied Statistics; Big Data Analysis; R and Complex Data Analytics; Database Principles

Publications

Peer-Reviewed Journal Articles

[1] Zhao, D., Luo, Q., Lai, X., **Sun, H.**, Qiu, Z., Zhang, H., & Sun, Z. (2026). Normalized dynamic fluorescence height: An alternative algorithm for chlorophyll *a* estimation in algae-dominated waters using hyperspectral remote sensing reflectance from in situ and spaceborne imagers. *Remote Sensing, 18*(14), 2332. https://doi.org/10.3390/rs18142332

Manuscripts in Preparation

[2] **Sun, H.**, Zhang, H., & Sun, Z. (in preparation). Nearshore morphology as a moderator of sea-level-rise effects on coastal water quality.

Research Experience

MSc Dissertation Jan 2026 – Aug 2026
Spatiotemporal Dynamics of Coastal Water Quality: The Hidden Role of Sea Level Rise *HKU*

- **Question:** Whether coastal morphology determines the direction of sea-level-rise effects on water quality in urbanised estuaries.
- **Data & methods:** 31 years of monthly data from 94 stations, combined with satellite altimetry, ocean and wave reanalysis, and coastal geophysical data; panel regression, causal mediation, and spatial autoregressive models in Python.
- **Findings:** A morphology-dependent sign reversal in the response across Hong Kong; stratification mediates 94% of the effect in enclosed embayments, robust to ENSO controls.

Multi-Satellite Chlorophyll-*a* Retrieval for Lake Taihu Sep 2025 – Dec 2025
Project Leader — Course GEOG7304, Satellite Data Processing and Information Extraction

- Led an 11-member team end to end: defined the research questions, planned the project, assigned tasks, and coordinated integration of results; awarded full marks.
- Designed a workflow harmonising Landsat 8, Sentinel-2, and GF-1 imagery to 30 m Landsat-equivalent reflectance, with in situ data for training and GOCI data for validation.
- Built a Random Forest retrieval model from spectral bands and water-quality indices that outperformed linear regression baselines, and produced multi-year, lake-wide chlorophyll-*a* maps.

Visiting Student, Ocean Colour Remote Sensing Sep 2024 – Jul 2025
School of Innovation and Entrepreneurship, SUSTech — Supervisor: Zhaohua Sun *Shenzhen, China*

- Trained in ocean colour science, marine optics, and the radiative properties of natural waters.
- Developed Python-based water quality retrieval algorithms from remote sensing reflectance and analysed the spatial distribution and temporal dynamics of water quality parameters.

Mathematical Contest in Modeling (MCM): Reimagine Masai Mara Feb 2023
Team project in remote sensing and environmental modelling

- Extracted land use, vegetation, and water information from remote sensing data for ecosystem modelling.
- Coupled ecosystem health and economic impact models through multi-objective optimisation to design and evaluate management strategies.

Professional Experience

Remote Sensing Engineer (Part-time) May 2026 – Aug 2026
AIEyes Technologies (Hong Kong) Co., Limited *Hong Kong SAR*

- Developed hyperspectral retrieval algorithm packages for 25+ water quality parameters (e.g., chlorophyll, CDOM, SPM), with an RMSE below 0.15.
- Contributed to a Hong Kong Science Park incubation program on AI-enabled water environment monitoring.

Data Engineer Assistant (Intern) Jul 2024 – Aug 2024
Anhua Ocean Intelligent Equipment (Shenzhen) Co., Ltd. *Shenzhen, China*

- Processed apparent optical property (AOP) data in Python to derive remote sensing reflectance (R_{rs}).
- Drafted requirements and webpage prototypes for a platform sharing AOP data with researchers.

Additional Training

Training Program in Machine Learning and NLP Apr 2024 – Jun 2024
Columbia University (online program)

- Applied machine learning and NLP (sentiment analysis, topic modelling) in Python to user reviews for location-based services.

Honors & Awards

Successful Participant, Mathematical Contest in Modeling (MCM) May 2024
Third-Class Scholarship, Xidian University Oct 2023
Provincial Third Prize, 14th "Zhengda Cup" National Market Research Competition Apr 2023

Skills

Programming	Python, R, MATLAB, Java, SQL (MySQL); intermediate C and LaTeX
Geospatial	ArcGIS, Google Earth Engine, GeoPandas
Statistics & ML	Random forest, panel regression, causal mediation analysis, spatial autoregressive models, clustering, NLP; SPSS
Languages	Chinese (native), English (fluent), German (intermediate)
Tests	IELTS 7.0 (Reading 8.0, Listening 7.5, Writing 6.5, Speaking 6.0), Jun 2024